SMT 和 NMT 训练尝试和初步比较

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1 SMT with Moses

1.1 Main Features of Moses[1]

Factored Translation Model

non-factored SMT noly deal with the surface form of words and has one phrase tabel

factored translation model the surface forms may be argmented with diffrent factors

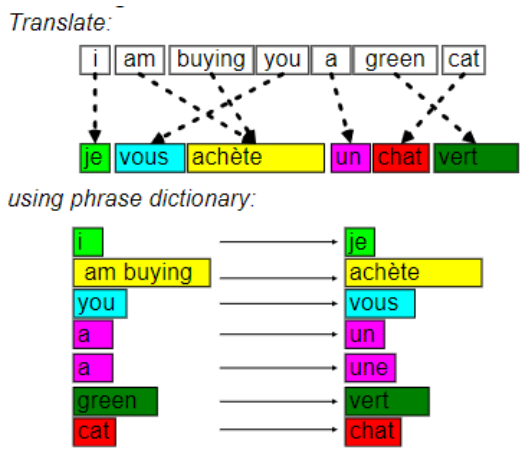

图 1: non-factored

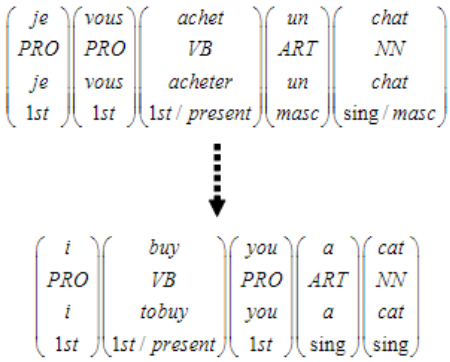

图 2: factored

Confusion Network Decoding

Efficient Data Structures for Translation Model and Language Models

1.2 Install Moses

Dependence Prepare

```
sudo apt-get install g++ git subversion libtool zlib1g-dev libicu-dev libboost-all-dev \
    libbz2-dev python-dev graphviz imagemagick make cmake libgoogle-perftools-dev \
    autoconf doxygen
```

Install Boost

```
cd /path/to/boost/
./bootstrap.sh
./b2 -j4 --prefix=$PWD --libdir=$PWD/lib64 --layout=system link=static install \
    || echo FAILURE
```

Install cmph

```
git clone https://github.com/zvelo/cmph.git
cd cmph
./configure
make
make install
```

Install GIZA

```
git clone https://github.com/moses-smt/giza-pp.git
cd giza-pp
make
cd ~/mosesdecoder
mkdir tools
cp ~/giza-pp/GIZA++-v2/GIZA++ ~/giza-pp/GIZA++-v2/snt2cooc.out \
   ~/giza-pp/mkcls-v2/mkcls tools
```

Install Moses

```
./bjam --with-boost=/path/to/boost --with-cmph=/path/to/cmph \
    --with-irstlm=/home/s0565741/workspace/irstlm-code \
```

```
    --with-xmlrpc-c=/home/s0565741/workspace/xmlrpc-c/xmlrpc-c-1.33.17 -8
```

1.3 Corpus prepare

1.3.1 Downlaod Corpus

```
cd smt
mkdir corpus
cd corpus
wget http://www.statmt.org/wmt13/training-parallel-nc-v8.tgz
tar zxvf training-parallel-nc-v8.tgz
```

1.3.2 Prepare the Data

tokenization 分词

```
~/mosesdecoder/scripts/tokenizer/tokenizer.perl -l en \
    < ~/corpus/training/news-commentary-v8.fr-en.en \
    > ~/corpus/news-commentary-v8.fr-en.tok.en
~/mosesdecoder/scripts/tokenizer/tokenizer.perl -l fr \
    < ~/corpus/training/news-commentary-v8.fr-en.fr \
    > ~/corpus/news-commentary-v8.fr-en.tok.fr
```

truecasting 大小写

```
# training truecast model first
~/mosesdecoder/scripts/recaser/train-truecaser.perl \
      --model ~/corpus/truecase-model.en --corpus \
      ~/corpus/news-commentary-v8.fr-en.tok.en
~/mosesdecoder/scripts/recaser/train-truecaser.perl \
```

```
      --model ~/corpus/truecase-model.fr --corpus    \
      ~/corpus/news-commentary-v8.fr-en.tok.fr


# truecasting
 ~/mosesdecoder/scripts/recaser/truecase.perl  \
   --model ~/corpus/truecase-model.en          \
   < ~/corpus/news-commentary-v8.fr-en.tok.en \
   > ~/corpus/news-commentary-v8.fr-en.true.en
 ~/mosesdecoder/scripts/recaser/truecase.perl  \
   --model ~/corpus/truecase-model.fr          \
   < ~/corpus/news-commentary-v8.fr-en.tok.fr \
   > ~/corpus/news-commentary-v8.fr-en.true.fr
```

cleaning 数据清洗

```
# limiting sentence length to 80
 ~/mosesdecoder/scripts/training/clean-corpus-n.perl \
     ~/corpus/news-commentary-v8.fr-en.true fr en \
     ~/corpus/news-commentary-v8.fr-en.clean 1 80
```

1.4 Language Model Training

```
mkdir ~/lm
cd ~/lm
~/mosesdecoder/bin/lmplz -o 3 \
     < ~/corpus/news-commentary-v8.fr-en.true.en > news-commentary-v8.fr-en.arpa.en
# -cores 设置使用核心数

#binarise (for faster loading)
```

```
~/mosesdecoder/bin/build_binary \
 news-commentary-v8.fr-en.arpa.en \
 news-commentary-v8.fr-en.blm.en
```

check the langauge model

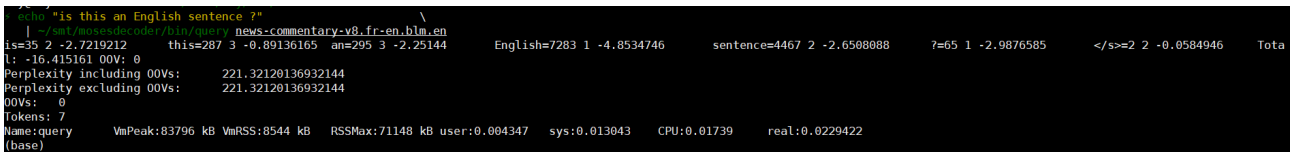

图 3: language-model-check

1.5 Training the Tranlation System

```
mkdir ~/working
cd ~/working
nohup nice ~/mosesdecoder/scripts/training/train-model.perl -root-dir train \
    -corpus ~/corpus/news-commentary-v8.fr-en.clean                          \
    -f fr -e en -alignment grow-diag-final-and -reordering msd-bidirectional-fe \
    -lm 0:3:$HOME/lm/news-commentary-v8.fr-en.blm.en:8                       \
    -external-bin-dir ~/mosesdecoder/tools >& training.out &
# --decoder-flags="-threads 8" 设置使用线程
```

1.6 Tuning

```
# tuning set
cd ~/corpus
wget http://www.statmt.org/wmt12/dev.
tgz
tar zxvf dev.tgz

# tokenisation & truecast
```

```
cd ~/corpus
~/mosesdecoder/scripts/tokenizer/tokenizer.perl -l en \
  < dev/news-test2008.en > news-test2008.tok.en
~/mosesdecoder/scripts/tokenizer/tokenizer.perl -l fr \
  < dev/news-test2008.fr > news-test2008.tok.fr
~/mosesdecoder/scripts/recaser/truecase.perl --model truecase-model.en \
  < news-test2008.tok.en > news-test2008.true.en
~/mosesdecoder/scripts/recaser/truecase.perl --model truecase-model.fr \
  < news-test2008.tok.fr > news-test2008.true.fr

# launch the tuning process
cd ~/working
nohup nice ~/mosesdecoder/scripts/training/mert-moses.pl \
 ~/corpus/news-test2008.true.fr ~/corpus/news-test2008.true.en \
 ~/mosesdecoder/bin/moses train/model/moses.ini --mertdir \
 ~/mosesdecoder/bin/ --decoder-flags="threads 8"\
 &> mert.out &
```

1.7 Tesing

```
# run
~/mosesdecoder/bin/moses -f ~/working/mert-work/moses.ini
```

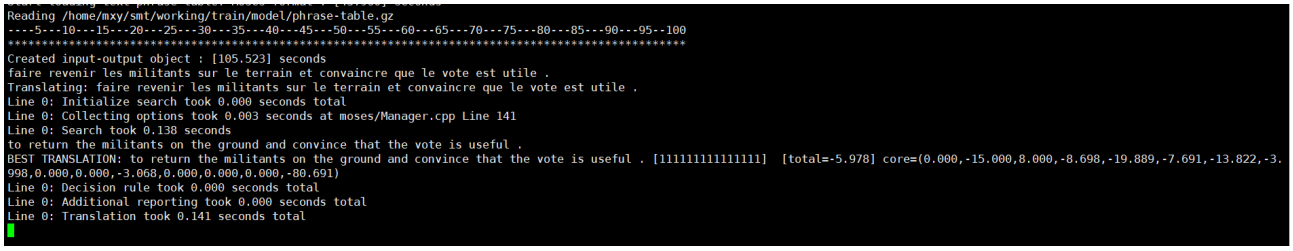

图 4: MosesTest

input:faire revenir les militants sur le terrain et convaincre que le vote est utile .

output:to return the militants on the ground and convince that the vote is useful .

google translate :bring activists back to the field and convince them that the vote is useful.

```
# take newstest2011 as test set ,so tokenisation and turecast this
cd ~/corpus
~/mosesdecoder/scripts/tokenizer/tokenizer.perl -l en \
  < dev/newstest2011.en > newstest2011.tok.en
~/mosesdecoder/scripts/tokenizer/tokenizer.perl -l fr \
  < dev/newstest2011.fr > newstest2011.tok.fr
~/mosesdecoder/scripts/recaser/truecase.perl --model truecase-model.en \
  < newstest2011.tok.en > newstest2011.true.en
~/mosesdecoder/scripts/recaser/truecase.perl --model truecase-model.fr \
  < newstest2011.tok.fr > newstest2011.true.fr

# retain the entries needed translate the test set
   cd ~/working
~/mosesdecoder/scripts/training/filter-model-given-input.pl \
   filtered -newstest2011 mert-work/moses.ini ~/corpus/newstest2011.true.fr \
  -Binarizer ~/mosesdecoder/bin/processPhraseTableMin

# first translation the test set & BLEU measure
   nohup nice ~/mosesdecoder/bin/moses \
  -f ~/working/filtered-newstest2011/moses.ini \
  < ~/corpus/newstest2011.true.fr \
  > ~/working/newstest2011.translated.en \
  2> ~/working/newstest2011.out
~/mosesdecoder/scripts/generic/multi-bleu.perl \
  -lc ~/corpus/newstest2011.true.en \
  < ~/working/newstest2011.translated.en
```

BLEU result:

output: BLEU = 23.57, 60.1/29.9/16.9/10.1

(BP=1.000, ratio=1.015, hyp_len=75885, ref_len=74758)

1.8 reference

1. http://www2.statmt.org/moses/?n=Development.GetStarted
2. http://www2.statmt.org/moses/?n=Moses.Baseline
3. https://achrafothman.net/site/how-to-install-moses-statistical-machine-translation

2 NMT with Fairseq

2.1 Install Fairseq

2.1.1 requirement

1. PyTorch version >= 1.5.0
2. Python version >= 3.6
3. NVIDIA GPU and NCCL

2.1.2 Install PyTorch

look up the version you need in the url below

https://pytorch.org/get-started/locally/

for me ,i should downlaod pytorch using the ctl showing in the figure below

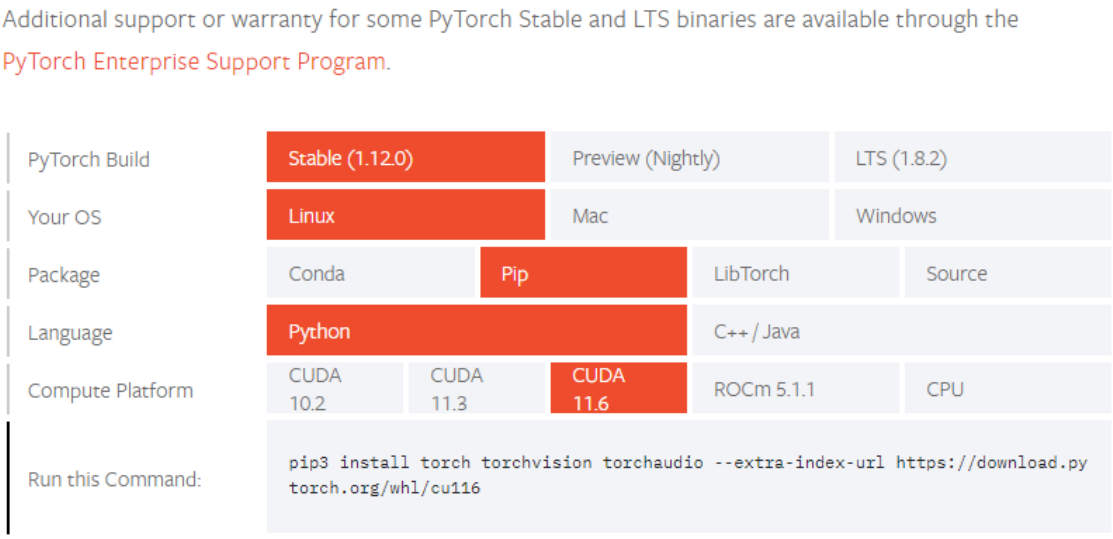

图 5: pytorch

2.1.3 Install Fairseq

```
git clone https://github.com/pytorch/fairseq
cd fairseq
pip install --editable ./
```

2.2 Data Prepare

```
cd examples/translation/
bash prepare-iwslt14.sh
cd ../..
TEXT=examples/translation/iwslt14.tokenized.de-en
fairseq-preprocess --source-lang de --target-lang en \
    --trainpref $TEXT/train --validpref $TEXT/valid --testpref $TEXT/test \
    --destdir data-bin/iwslt14.tokenized.de-en
```

2.3 Tarining

```
mkdir -p checkpoints/fconv
CUDA_VISIBLE_DEVICES=0 fairseq-train data-bin/iwslt14.tokenized.de-en \
  --optimizer nag --lr 0.25 --clip-norm 0.1 --dropout 0.2 --max-tokens 4000 \
  --arch fconv_iwslt_de_en --save-dir checkpoints/fconv
# --max-epoch 用来设定训练代数
```

2.4 Generate

```
fairseq -generate data-bin/iwslt14. tokenized .de-en \
    --path checkpoints/fconv/checkpoint_best.pt \
    --batch-size 128 --beam 5
```

2.5 Result

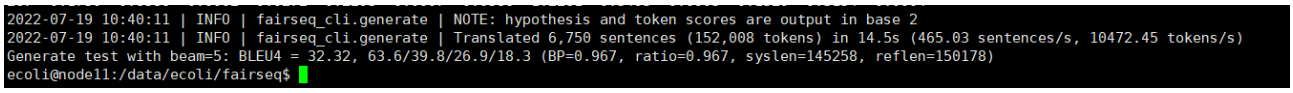

图 6: 6326CheckPoints

Generate test with beam=5: BLEU4 = 32.32, 63.6/39.8/26.9/18.3
(BP=0.967, ratio=0.967, syslen=145258, reflen=150178)

2.6 Reference

1. https://fairseq.readthedocs.io/en/latest/getting_started.html#training-a-new-model
2. https://zhuanlan.zhihu.com/p/448147465
3. http://www.linzehui.me/2019/01/28/%E7%A2%8E%E7%89%87%E7%9F%A5%E8%AF%86/%E5%A6%82%E4%BD%95%E4%BD%BF%E7%94%A8fairseq%E5%A4%8D%E7%8E%B0Transformer%20NMT/

3 Compare

3.1 Change the Data

Training a New Model in Moses first copy the data set in fairseq to Moses ,the path is /fairseq/examples/translation/iwslt14.tokenized.de-en the data were devided into three groups as train valid and test ,so we could corresponded these three sets of data to traning ,tuning and test stages. Use the same data as fairseq to train a new model so we can compare the result between SMT and NMT.

Then follow the guidance of Moses the new model has been trained and here's the result of BLEU.

BLEU = 11.44, 43.6/16.1/7.1/3.4

(BP=1.000, ratio=1.029, hyp_len=154505, ref_len=150178)

3.2 Compare With the Result of Fairseq

	BLEU	BP	ratio	hyp_len	ref_len
Moses	11.44	1.000	1.029	154505	150178
Fairseq	32.32	0.967	0.967	145258	150178

可以看出再相同的数据集下 SMT 的训练效果显著低于 NMT 的训练效果, 使用 compare mt 得到的比较数据,sys1 为 Moses 所得数据，sys2 为 Fairseq 所得数据

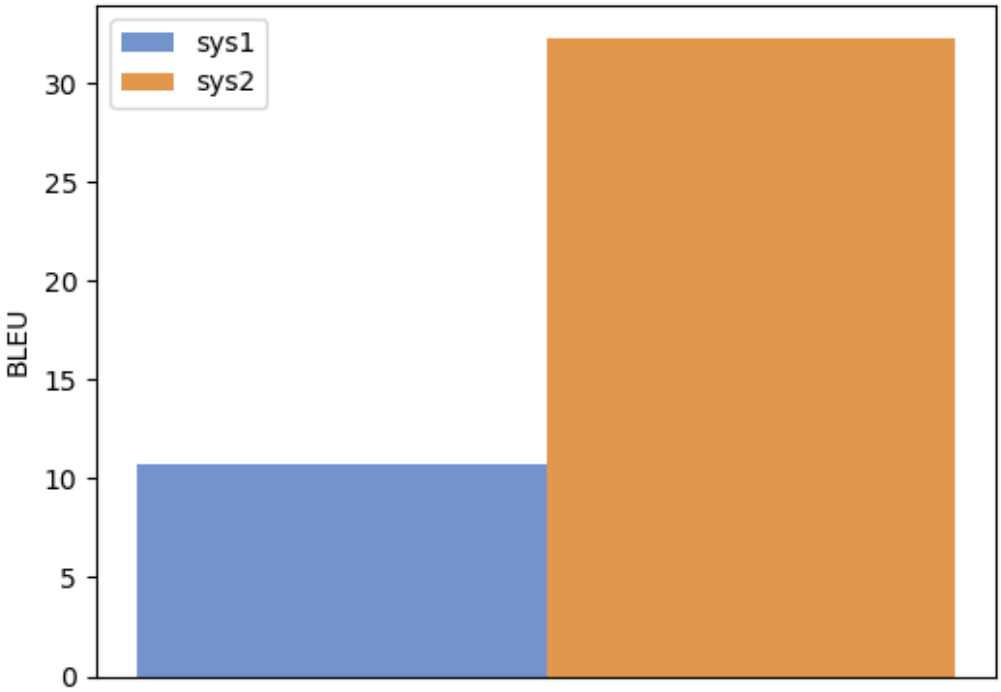

图 7: Score Comparison

frequency	sys1	sys2
<1	0.0000	0.0000
1	0.1270	0.3750
2	0.2261	0.4895
3	0.2522	0.4944
4	0.2581	0.5018
[5,10)	0.2400	0.5147
[10,100)	0.3002	0.5458
[100,1000)	0.4761	0.5953
>=1000	0.6722	0.7507

表 1: table 1

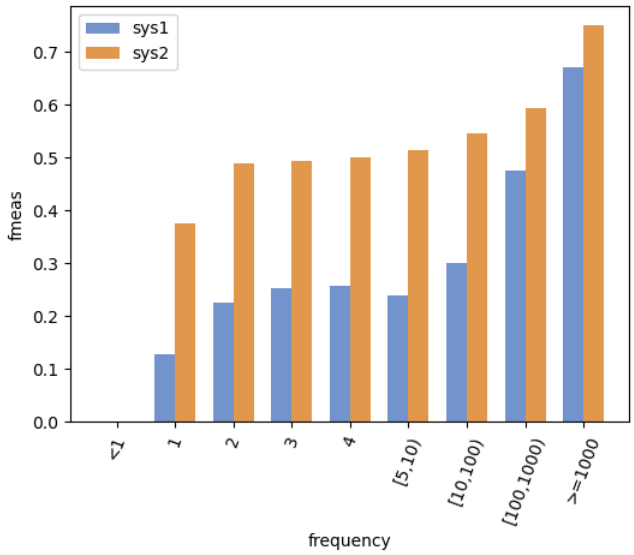

图 8: Word Accuracy Analysis

length	sys1	sys2
<10	10.5767	37.1071
[10,20)	11.0897	35.1189
[20,30)	10.3008	32.9503
[30,40)	10.5195	31.2436
[40,50)	10.6186	29.5829
[50,60)	10.5664	30.3212
>=60	11.9584	27.9565

表 2: table 2

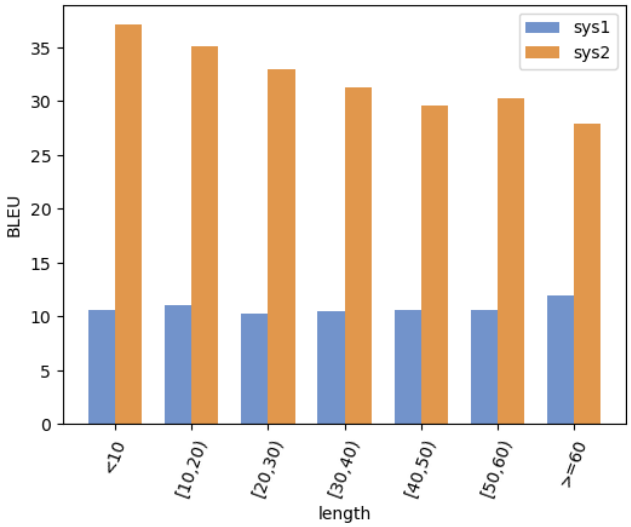

图 9: Sentence Bucket Analysis

lengthdiff	sys1	sys2
<-20	12	18
[-20,-10)	34	89
[-10,-5)	232	492
-5	144	264
-4	254	356
-3	340	498
-2	560	703
-1	790	934
0	1078	1328
1	963	758
2	681	483
3	490	274
4	357	180
5	247	103
[6,11)	453	204
[11,21)	98	41
>=21	17	25

表 3: table 3

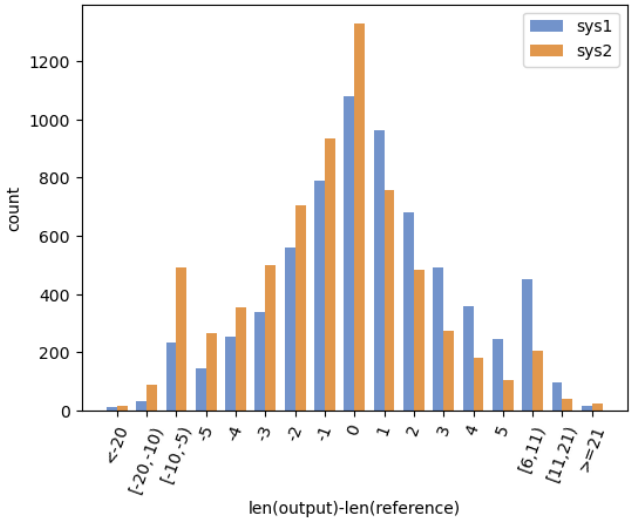

图 10: Sentence Bucket Analysis

sentbleu	sys1	sys2
<10.0	1772	222
[10.0,20.0)	3130	1245
[20.0,30.0)	1038	1476
[30.0,40.0)	456	1366
[40.0,50.0)	180	925
[50.0,60.0)	90	636
[60.0,70.0)	34	341
[70.0,80.0)	23	181
[80.0,90.0)	4	139
>=90.0	23	219

表 4: table 4

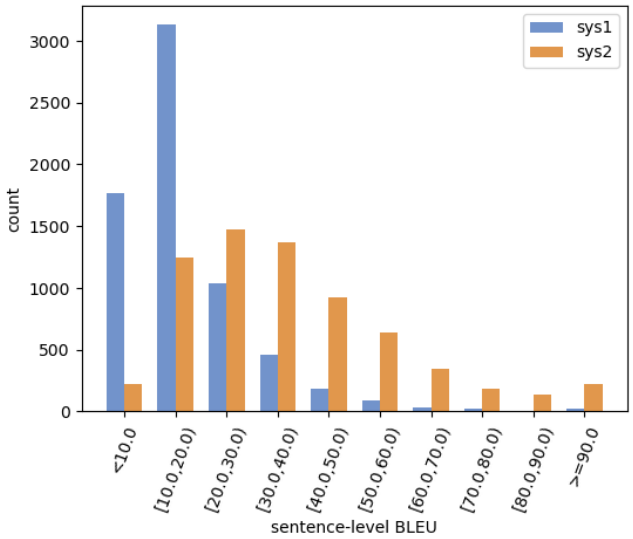

图 11: Sentence Bucket Analysis

3.3 sentences where sys1>sys2 at sentence-level BLEU

	Output	sentbleu
Ref	it is nat@@ anz , and it is only nat@@ anz .	
sys1	it is nat@@ anz , and it is only nat@@ anz .	100.0000
sys2	it 's nat@@ as@@ y , and it 's just nat@@ ching .	20.8304

表 5: table 5

	Output	sentbleu
Ref	pattern recogni@@ tion should also work .	
sys1	pattern recogni@@ tion should still work .	59.1546
sys2	so , there was a way of working .	12.1852

表 6: table 6

3.4 sentences where sys2>sys1 at sentence-level BLEU

	Output	sentbleu
Ref	so i want you to think about a thought experiment .	
sys1	I want to so that they have a ge@@ dan@@ ken@@ experiment .	15.2293
sys2	so i want you to think about a thought experiment .	100.0000

表 7: table 7

	Output	sentbleu
Ref	and that 's actually a conser@@ va@@ tive esti@@ mate .	
sys1	and this is in the did a konser@@ va@@ tive schä@@ tzung .	14.5507
sys2	and that 's actually a conser@@ va@@ tive esti@@ mate .	100.0000

表 8: table 8

3.5 Conclusion and Possible reasons for this result

结论：

1. 在对低词频词进行翻译时，NMT 的效果远优于 SMT 的翻译效果，对高频词的翻译中 SMT 也有不错的表现
2. NMT 在不同长度的句子翻译中都是远优于 SMT 但是 NMT 的翻译效果在句子长度增加时有下降的趋势
3. 两种翻译在翻译结果长度与参考结果长度差上都有类似正态分布
4. SMT 翻译的单句 BLEU 绝大部分分布在 0.2 以下以 0.1 0.2 区间为主要分布；NMT 翻译的单句 BLEU 集中在 0.1 0.5 区间中，较高得分区间 NMT 有较大潜力

可能的原因猜想：

1. SMT 和 NMT 在建立模型和训练算法上的不同

2. 在处理数据时 Moses 仅采用了 tokenization，truecasting，cleaning 三个步骤，而 fairseq 在这基础上采用了 BPE 算法

BLEU[2]

BLEU (BiLingual Evaluation Understudy) is a metric for automatically evaluating machine-translated text. The BLEU score is a number between zero and one that measures the similarity of the machine-translated text to a set of high quality reference translations. A value of 0 means that the machine-translated output has no overlap with the reference translation (low quality) while a value of 1 means there is perfect overlap with the reference translations (high quality). and there is a rough guideline

BLEU Score	Interpretation
< 10	Almost useless
10 - 19	Hard to get the gist
20 - 29	The gist is clear, but has significant grammatical errors
30 - 40	Understandable to good translations
40 - 50	High quality translations
50 - 60	Very high quality, adequate, and fluent translations
> 60	Quality often better than human

The following color gradient can be used as a general scale interpretation of the BLEU score:

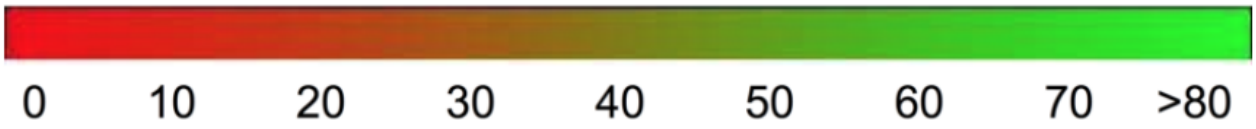

图 12: BLEU-guideline

参考文献

[1] Philipp Koehn, Hieu Hoang, Alexandra Birch, Chris Callison-Burch, Marcello Federico, Nicola Bertoldi, Brooke Cowan, Wade Shen, Christine Moran, Richard Zens, et al. Moses: Open source toolkit for statistical machine translation. In *Proceedings of the 45th annual meeting of the association for computational linguistics companion volume proceedings of the demo and poster sessions*, pages 177–180, 2007.

[2] Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. Bleu: a method for automatic evaluation of machine translation. In *Proceedings of the 40th annual meeting of the Association for Computational Linguistics*, pages 311–318, 2002.